Lectures

Stat 154/254: Statistical Machine Learning

Lectures marked with $\star$ are still in progress.



Introduction

- Class introduction
- Key concepts

Unit 1: Regression in a machine learning setting

- Linear regression in matrix notation (review)
- Linear regression in an ML setting
- Basis expansions and splines for linear regression
- L1 and L2 penalization

Unit 2: Classification

- Empirical risk minimization for classification
  - Perceptron algorithm **(extra, not covered in class)**
- Trading off false positives and negatives

Unit 3: Risk and complexity

- Regret decomposition
- Asymptotics and bias–variance tradeoff in linear regression
- Generalization bounds
  - Generalization error and uniform laws for smooth function classes
  - VC dimension and zero–one loss
- Cross validation
  - Held-out data and cross-validation
  - Simulated bias–variance tradeoff example
  - Cross–validation cautionary note

Unit 4: Trees and weak learners

- Classification and regression trees
- Bagging weak learners
- Boosting weak learners

Unit 5: Neural networks and optimization

This unit will mostly not have its own notes, and will instead be based on readings from Prince (2023), which is freely available online.

- Black-box optimization
- Shallow and deep neural networks (Prince (2023) chapters 3 and 4)
  - Visualizations
- Optimization and generalization in neural networks (Prince (2023) chapters, 6, 9)
- Architecture as structure (Prince (2023) chapters 10, 11)
- Simulation-based inference and TabPFN

Prerequisites

Mathematical statistics, probability, linear algebra, multivariate calculus, and Python are all prerequisites.

For assistance with mathematics prerequisites, the Student Learning Center provides additional assistance.

A summary of the required linear algebra skills can be found here.

Prince, Simon J. D. 2023. *Understanding Deep Learning.* The MIT Press. http://udlbook.com.